INCH-POUND

MIL-PRF-1/1773A
1 December 1997
SUPERSEDING
MIL-E-1/1773
21 July 1989

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON
TYPE DOD-042

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Coaxial fixed frequency, 9,375 MHz ± 30 MHz, pulse type, 65-kw rated peak-power output, integral permanent magnet, air cooled.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V	If A	tk sec	tpc µs	rrv kv/µs	Pi W	ib a	VSWR	Alt ft
Maximum Minimum	14 ----	10 (surge) ----	---- 90	5.5 0.4	187 54	240 ----	15 11	1.5:1 ----	10,000 ----

ABSOLUTE RATINGS:

Parameter: Unit:	Anode T °C	Cathode bushing T °C	Storage ambient T °C
Maximum Minimum	150 ---- (See note 17)	175 -55 (See note 15)	+85 -55

PHYSICAL CHARACTERISTICS:

See figures 1, 1A, 2, 2A, 3, 3A, 4, and 4A.
Weight: 6 pounds (approximate).

TEST CONDITIONS:

Parameter: Unit: Tolerance:	Ef V ---- 3/	tpc µs ±10%	Du ---- ----	rrv kv/µs ±10%	Ib mA dc ----
Test 1:	9.1	0.4	0.00065	170	9.8
Test 2:	9.1	0.4	0.00065	170	----
Test 3:	7.9	5.0	0.001	110	15
Test 4:	7.9	5.0	0.001	110	----
Test 5:	7.9	5.0	0.001	60	----

GENERAL:

Qualification - Required.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Test	Conditions	Symbol	Limits		Unit
						Min	Max	
Qualification inspection								
Life test	----	10/ 17/	----	VSWR = 1.5:1 (min) with phase varying through a minimum of 0.5 λ approximately every 15 minutes	----	2,000	----	Hours
Life-test end points:								
Power levels	4250	1/	3		Po	56	----	W
Frequency	----	----	3		F	9,345	9,405	MHz
Spectrum	4308	4/ 5/ 12/	3		BW	----	2.5/tpc	MHz
Spectrum	4308	4/ 5/ 12/	3		Ratio	8	----	dB
Pulse stability	4315	12/ 12/	1 3		MP MP	---- ----	2.0 2.0	% %
End of life-test end-points:								
Vibration, mechanical	1032	13/	1	10 G; f = 50 Hz to 500 Hz; varied over complete range in not less than 5 minutes at uniform rate.	ΔF	----	±0.5	MHz
Direct-interelectrode capacitance	4266	----	----	Measured between cathode terminal and mounting plate.	Cin	9	14	pF
Barometric pressure, reduced	4028	11/	3	Pressure = 55 mm Hg absolute	----	----	----	----
Conformance Inspection, part 1								
Heater or filament current, nonoperating	4289	----	----	Ef = 12.6 V; tk = 180 seconds (min)	If	2.0	2.4	A
Pressurizing	4003	----	----	40 to 45 psia, input and output assemblies.	----	----	----	----
Power levels	4250	1/ 1/	1 3		Po Po	44 65	---- ----	W W
Spectrum	4308	4/ 5/ 15/ 4/ 5/ 16/	2 4		BW BW	---- ----	2.0/tpc 2.5/tpc	MHz MHz
Spectrum	4308	4/ 5/ 15/ 4/ 5/ 16/	2 4		Ratio Ratio	10 8	---- ----	dB dB
Pulse voltage	4306	----	3		epy	14.7	14.9	kv

See notes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Test	Conditions	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1 - Continued								
Pulse stability	4315	---- 9/	1 3		MP MP	---- ----	0.25 0.25	% %
Frequency pulling figure	4310	----	1		ΔF	----	5	MHz
Frequency	----	17 on figure 1	3	Anode T at reference point = +100°C ±10°C	F	9,345	9,405	MHz
Frequency pushing figure	4311	----	3	ib = 12 a to 15 a	ΔF/Δib	----	0.1	MHz/a
Conformance inspection, part 2								
Vibration	----	13/	----	Ef = 12.6 V; 15 G; F = 60 Hz; duration 15 minutes; no heater cathode shorts during test	----	----	----	----
Shelf life	----	6/	----	t = 90 days	----	----	----	----
Shelf life end points:								
Power output	----	----	3		Po	65	----	W
Stability	----	----	3		MP	----	0.25	%
Pulse stability	4315	---- ----	3 5		MP MP	---- ----	0.5 0.25	% %
Conformance inspection, part 3								
Life test provisions								
Intermittent life	1501	10/ 14/	---- ----	Group D; VSWR = 1.5:1 (min) with phase varying through a minimum of 0.5λ approximately every 15 minutes	----	2,000	----	Cycles
Life-test end points:								
Power levels	4250	1/	3		Po	56	----	W
Frequency	----	----	3		F	9,345	9,405	MHz
Spectrum	4308	4/ 5/ 12/	3		BW	----	2.5/tpc	MHz
Spectrum	4308	4/ 5/ 12/	3		Ratio	8.0	----	dB
Pulse stability	4315	12/ 12/	1 3		MP MP	---- ----	2.0 2.0	% %

See notes at end of table

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Test	Conditions	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3 -Continued								
End of life-test end points periodic tests								
Shock, specified pulse	1042	2/	2	50 G; 4 ms duration, no voltages	----	----	----	----
Magnetron nonoscil-lating impedance (phase of sink)	4309	7	----		Distance	0.26	0.4	λg
Temperature coefficient	4027	q	1	Anode = +70°C to +100°C at reference point	ΔF/°C	----	0.25	MHz/°C
High-and-low-temperature per-formance (operating)	1047	----	1	tk = 90 s max; T = -55°C	MP	----	1.0	%
Forced convection	1143	8	1		ΔT	----	50	°C

1/ The tube heater shall be protected against arcing by the use of a capacitor that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.

2/ The magnetron shall be capable of specified performance when subjected to a 50 G, 4 ms duration, half sine shock pulse in accordance with MIL-STD-202, method 213B.

The mounting plate of the magnetron shall be bolted to a test fixture that is bolted directly to the table of the shock test machine.

The magnetron shall be given one shock pulse in each of the following directions:

a. Parallel to cathode, with cathode terminals pointing away from the impact surface.

b. Perpendicular to cathode axis and output waveguide axis.

c. Perpendicular to cathode axis and parallel to the output waveguide axis.

Post shock test results shall show no mechanical failure and the magnetron shall meet the electrical requirements of QCI, part 1.

3/ A tk of 180 seconds (minimum) with an Ef of 12.6 volts is required before test.

4/ The tube shall be operated into a transmission line with a VSWR of 1.5:1 adjusted in phase to produce maximum spectrum degradation.

5/ A suitable spectrum shall be considered one in which the major lobe has a shape such that its slope does not change sign more than once for power levels greater than the specified dB below its peak.

6/ This test shall be performed on three tubes per month with zero failures when tube is in continuous production, but shipments of that month's production shall not be held pending completion of the test. If this condition is not met, shipments shall be halted until three tubes of current production are tested with zero failures.

7/ Using a standard cold-test technique, the phase of sink as measured from the output flange to the first minimum, toward the load, shall be within the limits specified.

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TABLE I. Testing and inspection - Continued.

8/ An airflow of 15 cfm at approximately 760 mm Hg shall be directed on the cooling fins from an orifice of 2.5 inches (63.5 mm) x 1.188 inches (30.18 mm). The temperature rise shall be measured at that point on the anode block as shown on figure 1 (see figure 1, note 17).
9/ This test shall be the first one performed after the specified holding period.
10/ Air cooling shall be adjusted so that the anode block runs at +150°C, or at the maximum temperature it will reach in the absence of cooling, whichever is lower. This shall be adjusted during the test condition 3 portions of the cycle.
11/ The tube shall be operated in a transmission line with a load VSWR of 1.5:1 and a variable phase. At specified pressure, there shall be no evidence of breakdown at the input or output assemblies.
12/ If during life test the tube does not meet the specified limits, it shall be recycled for an additional five cycles. At such time, the test shall be repeated. Should the tube fail the second test, it shall be considered unsatisfactory.
13/ Direction of vibration shall be in a plane perpendicular to the axis of the cathode.
14/ Life test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration
Standby	----	12.6 V	3 minutes
Test 1	9.8 mA	9.1 V	3 minutes
Test 3	15.0 mA	7.9 V	15 minutes
Off	----	0	9 minutes (minimum)

15/ Measurements shall be made at 11 a and 15 a.
16/ Measurements shall be made at 12 a and 15 a.
17/ Life test shall be conducted with the following cycle:

Condition	Ib	Ef	Duration
Standby	----	12.6 V	3 minutes
Test 1	9.8 mA	9.1 V	15 minutes
Test 3	15.0 mA	7.9 V	30 minutes
Off	----	0	12 minutes

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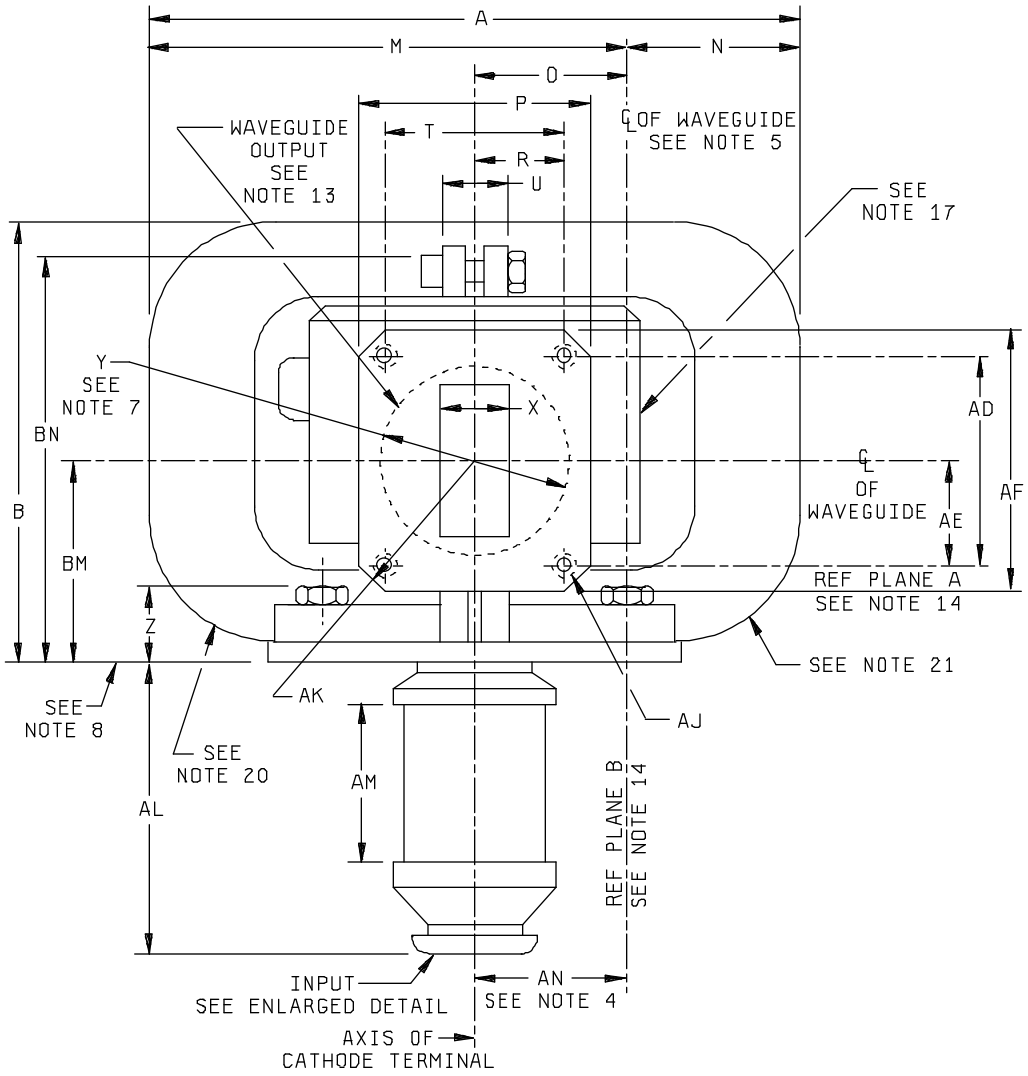


FIGURE 1. Outline drawing of electron tube type DOD-042 (front view).

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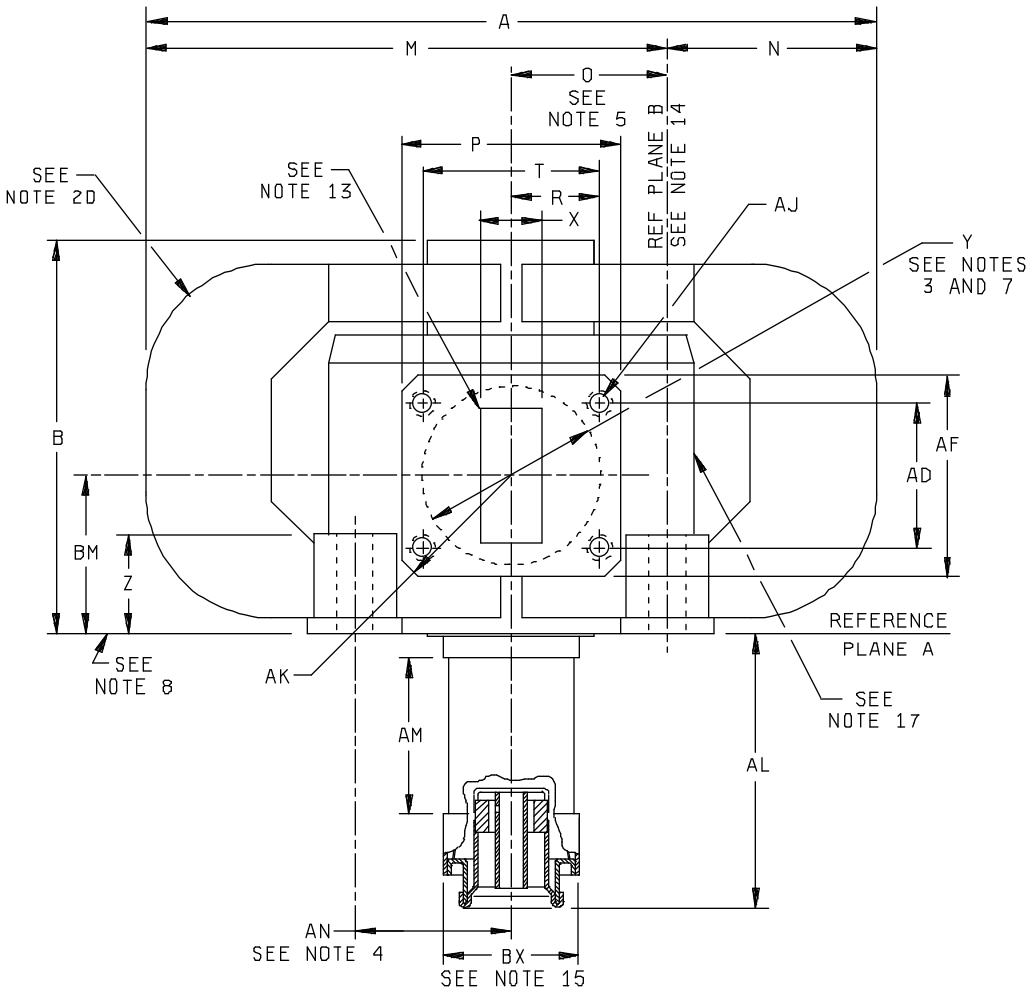


FIGURE 1A. Alternate outline drawing of electron tube type DOD-042 (front view).

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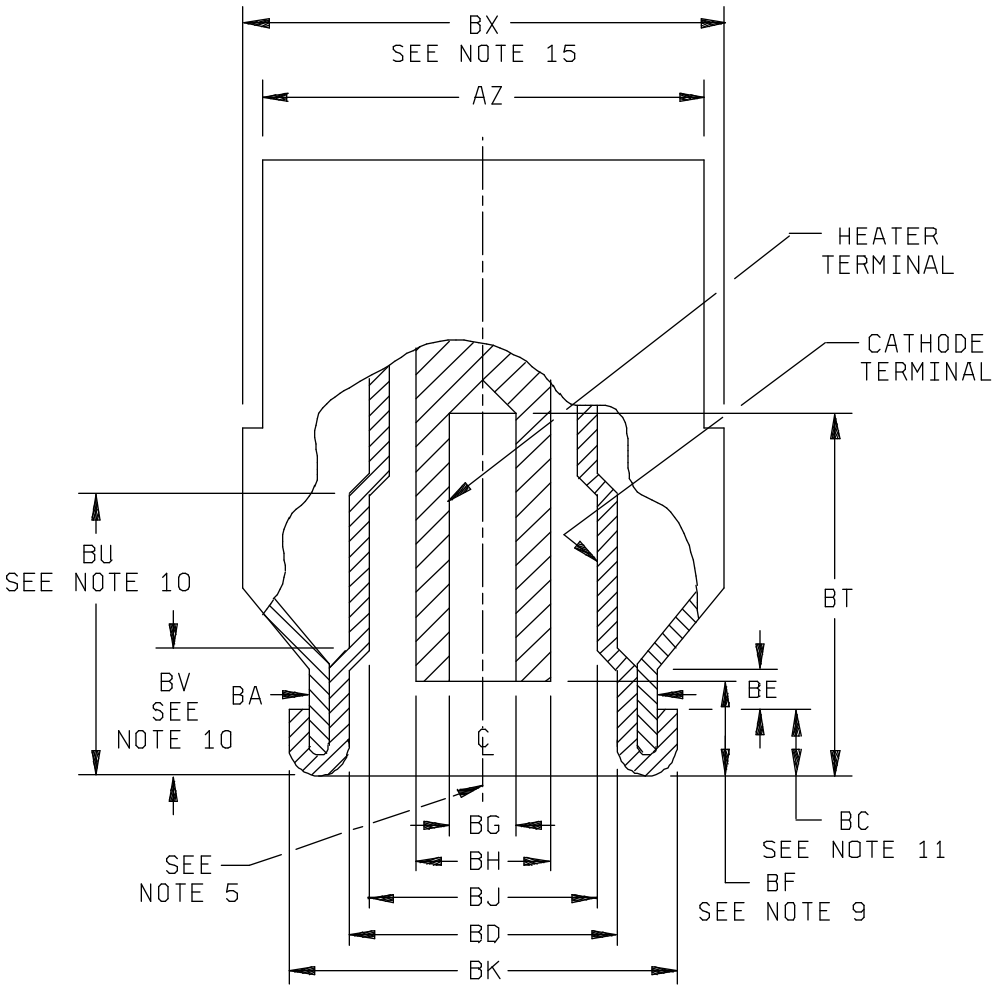


FIGURE 2. Outline drawing of electron tube type DOD-042 (detail enlarged).

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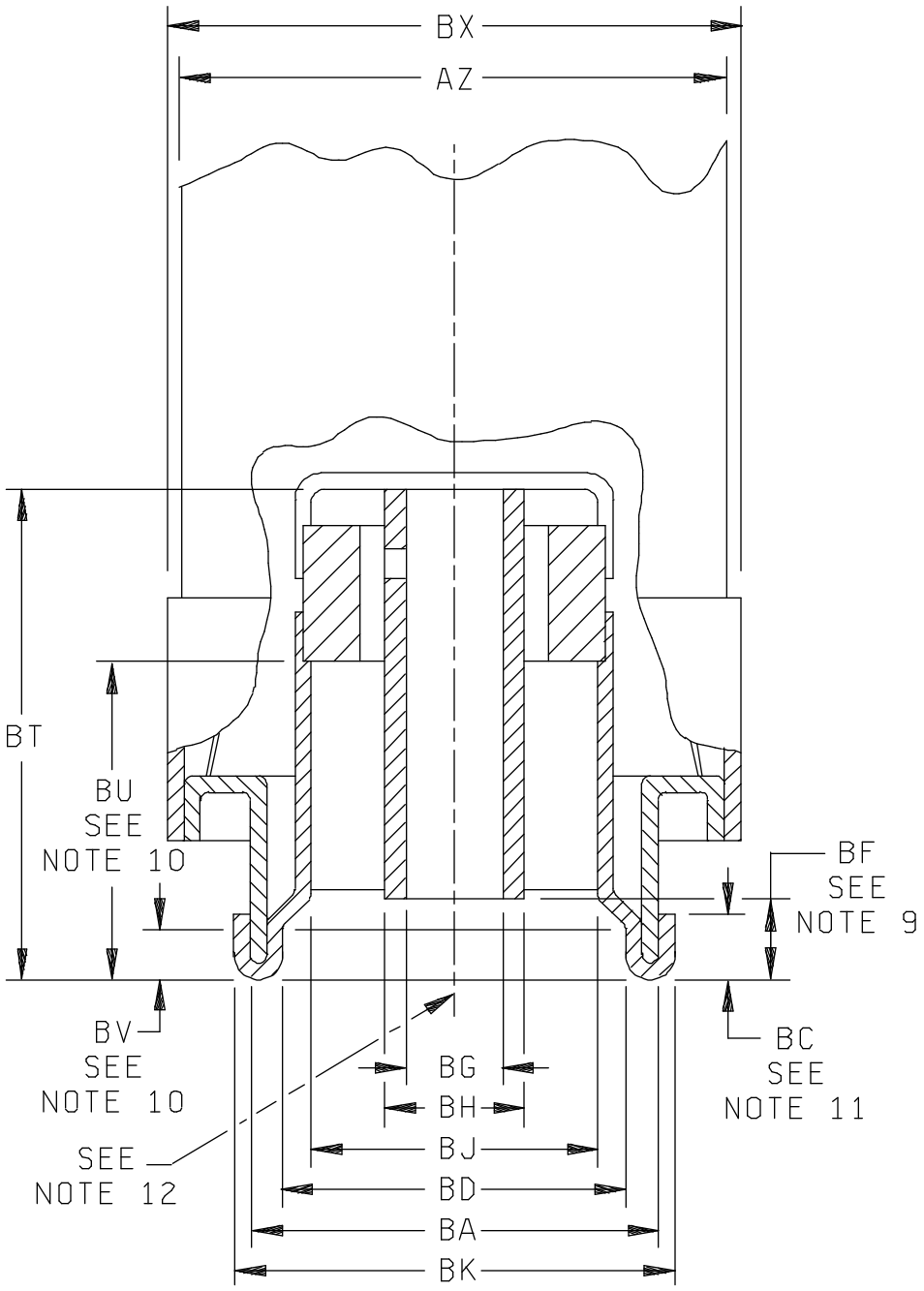


FIGURE 2A. Alternate outline drawing of electron tube type DOD-042 (detail enlarged).

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Figure	Ltr	Dimensions				Notes
		Inches		Millimeters		
Conformance inspection, part 1						
		Min	Max	Min	Max	
1, 1A	O	1.250	1.280	31.75	32.51	5
1, 1A	R	.732	.742	18.59	18.85	
1, 1A	T	1.470	1.478	37.34	37.54	
1, 1A	Z	.594	.656	15.09	16.66	
1, 1A	AD	1.348	1.356	32.24	34.44	
1	AE	.671	.681	17.04	17.30	
1, 1A	AL	2.094	2.219	53.19	56.36	
2, 2A	BC	.115	.135	2.92	3.43	11
2, 2A	BD	.610	.650	15.49	16.51	
2, 2A	BF	.125	.187	3.18	4.75	9
2, 2A	BG	.164	.174	4.17	4.42	
2, 2A	BH	.234	.266	5.94	6.76	
2, 2A	BJ	.532	.545	13.51	13.84	
2, 2A	BK	.825	.838	20.96	21.29	
1, 1A	BM	1.386	1.426	35.20	36.22	
Conformance inspection, part 2						
1, 1A	A	----	5.938	----	150.83	
1, 1A	B	----	3.156	----	80.16	
1, 1A	M	---	4.250	----	107.95	
1, 1A	N	----	1.688	----	42.88	
1	U	----	1.438	----	36.53	
1, 1A	AM	1.250	----	31.75	----	
2	BE	.125	----	3.18	----	
1	BN	----	2.938	----	74.63	
2, 2A	BT	.750	----	19.05	----	
2, 2A	BU	.516	----	13.11	----	10
2, 2A	BV	----	.200	----	5.08	10
Conformance inspection, part 3						1
1, 1A	P	----	1.830	----	46.48	
1, 1A	AF	----	1.830	----	46.48	
Reference dimensions						
1, 1A	X	0.497		12.62		
1 1A	Y	1.625 1.516		41.28 38.51		
1, 1A	AJ	.164-32 UNC-2B				
1, 1A	AK	1.156		29.36		
1, 1A	AN	1.265		32.13		4
2 2A	AZ	1.125 1.100		28.58 27.94		
2 2A	BA	0.750 0.772		19.05 19.61		
2 2A	BX	1.000 1.125		25.40 28.57		15

FIGURES 1, 1A, 2, and 2A. Outline drawing of electron tube type DOD-042.

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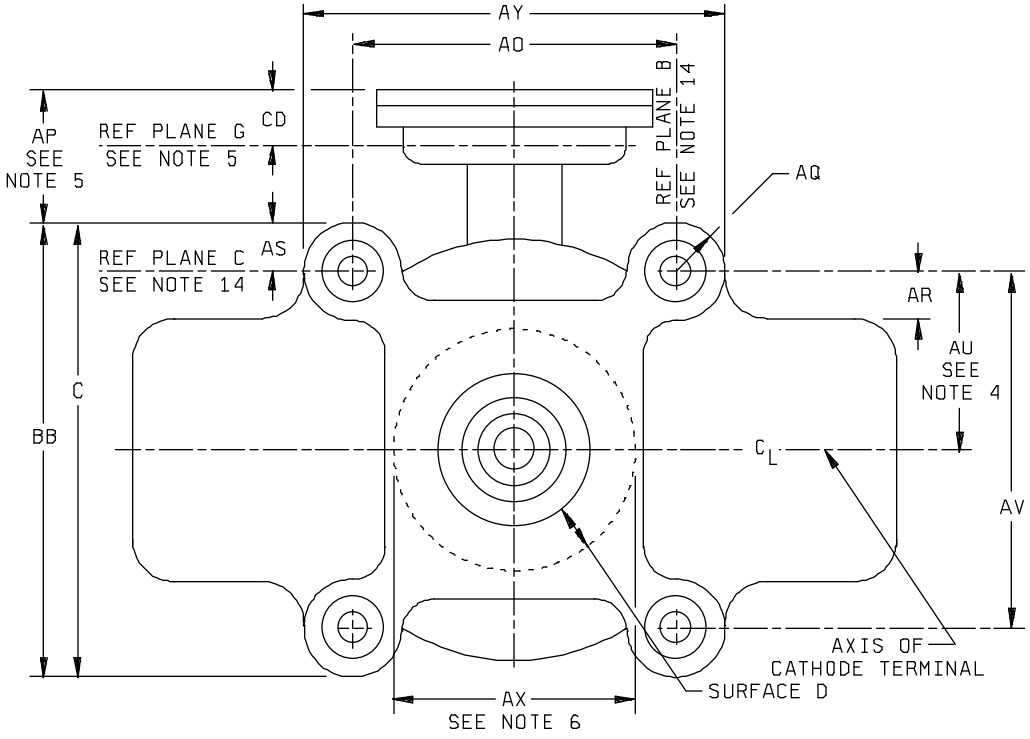


FIGURE 3. Outline drawing of electron tube type DOD-042 (bottom view).

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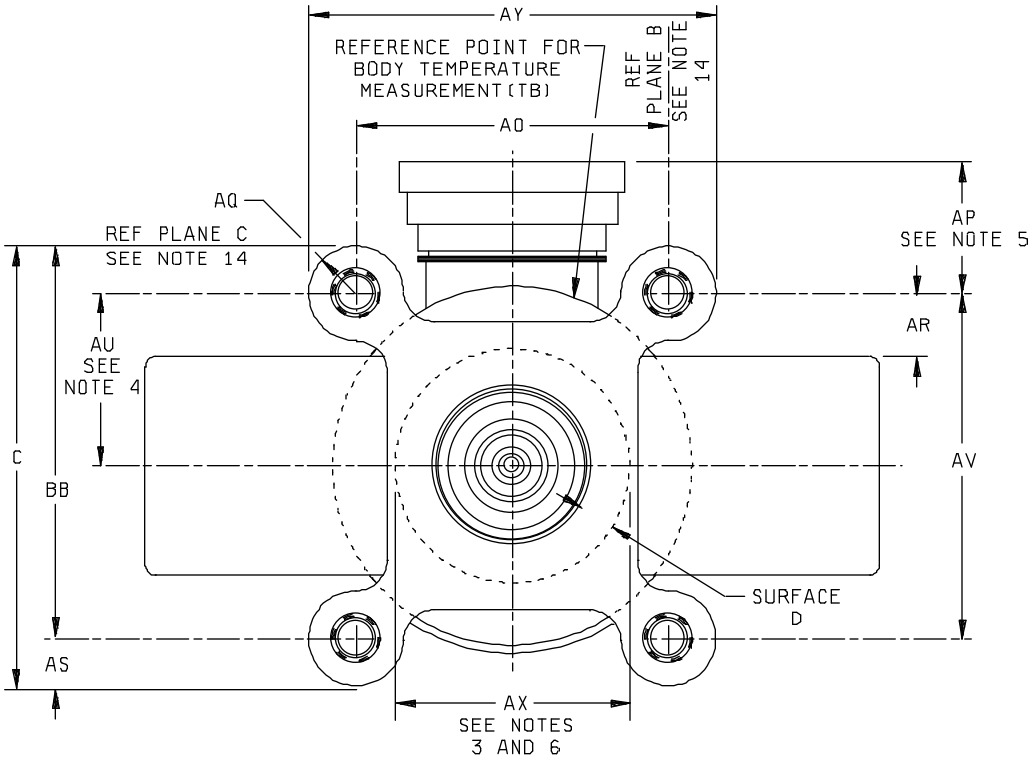


FIGURE 3A. Alternate outline drawing of electron tube type DOD-042 (bottom view).

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Figure	Ltr	Dimensions				Notes
		Inches		Millimeters		
Conformance inspection, part 1						
		Min	Max	Min	Max	
3, 3A	AP	.990	1.040	25.15	26.42	5
3, 3A	AV	2.771	2.791	70.38	70.89	
3, 3A	AO	2.521	2.541	64.03	64.54	
Conformance inspection, part 2						
3, 3A	C	----	3.625	----	92.08	
3, 3A	AR	----	.594	----	15.09	
3, 3A	AS	----	.422	----	10.72	
3, 3A	AX	1.875	----	47.62	----	6
3, 3A	AY	----	3.483	----	87.33	
3, 3A	BB	----	3.203	----	81.36	
Conformance inspection, part 3 periodic check						1
Reference dimensions						
3, 3A	AQ	.375		9.52		
3, 3A	AU	1.391		35.33		4
3	CD	.411		10.44		5

FIGURES 3 and 3A. Outline drawing of electron tube type DOD-042 (bottom view).

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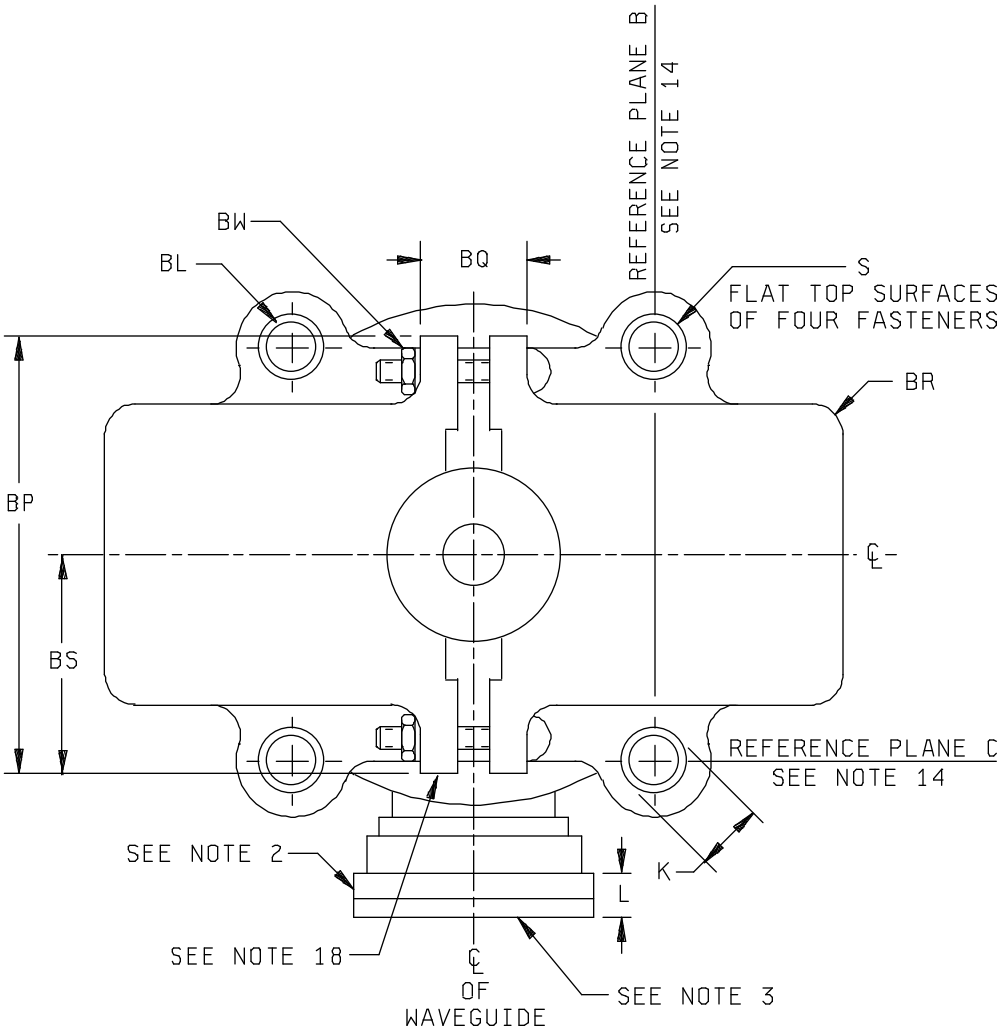


FIGURE 4. Outline drawing of electron tube type DOD-042 (top view).

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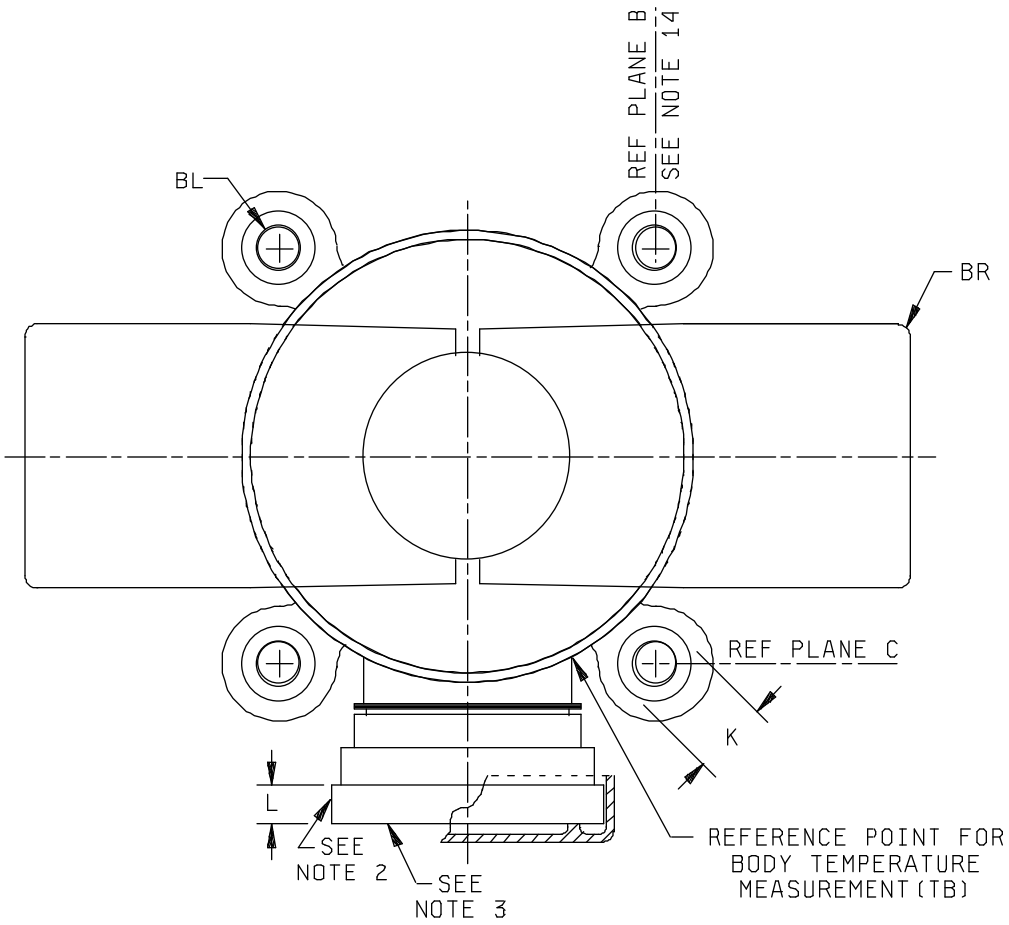


FIGURE 4A. Alternate outline drawing of electron tube type DOD-042 (top view).

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Figure	Ltr	Dimensions				Notes
		Inches		Millimeters		
Conformance inspection, part 1						
		Min	Max	Min	Max	
4	BS	1.334	1.438	33.88	36.53	
4, 4A	BL	.276	.286	7.01	7.26	
Conformance inspection, part 3						
4	BP	----	2.875	----	73.02	
4	BQ	----	.625	----	15.88	
4, 4A	BR	.031R	----	0.79	----	
Conformance inspection, part 2						
Reference dimensions						
Nominal						
4, 4A	K	.500		12.70		
4, 4A	L	.250		6.35		
4	BW	.138-32 UNC-2A				

NOTES:

1. If these tests have been previously performed by the tube manufacturer, and if the material, manufacturing process, and test requirements are the same as when the tests were performed, the subject test(s) need not be performed.
2. All metal surfaces black finished except those designated surface S and surface D. Surfaces S shall be gold, silver, nickel or bright alloy plated, or brass surfaces, except as specified.
3. Hermetic connections may be made to surface D.
4. Axis of cathode terminal shall occupy specified location within .047 inch (1.19 mm) radius. Note 7 shall apply.
5. Limits include angular and lateral deviations.
6. With diameter AX resting on a plane surface coincident with reference plane A, a gauge .010 inch (0.25 mm) thick and .125 inch (3.18 mm) wide shall not enter, and areas of the base plate outside diameter AX shall be within .010 inch (0.25 mm) of the plane surface.
7. With diameter Y resting on a plane surface, a gauge .005 inch (0.13 mm) thick and .125 inch (3.18 mm) wide shall not enter.
8. Any portion of the assembly which extends below reference plane A shall be within .625 inch (15.88 mm) radius of specified axis of input.
9. Defines extremity of cylindrical section designated BG.
10. Defines extremity of cylindrical section designated BJ.
11. Clamping shall be confined to this area.
12. Heater terminal and cathode terminal shall be concentric within .010 inch (0.25 mm).
13. Protective covers shall be provided with tubes.
14. Reference planes A, B, and C are mutually perpendicular.
15. Reference point for cathode temperature measurements. Located on the diameter, dimension BX.
16. Gold, silver, or nickel plate over full extent of maximum plug penetration.
17. Reference point for anode temperature measurements. Located on body near output section.
18. Extremity of magnet lug shall coincide with reference plane C within .094 inch (2.39 mm).
19. For vibration tests, planes of vibration shall be reference planes A, B, and C.
20. See note 21. Warning: Maintain minimum 2.00 inches (50.8 mm) between this magnet and magnetic materials (magnets, steel tool, and plates).
21. Manufacturers code designation and tube serial number shall appear within this area.

FIGURES 1 THROUGH 4A. Outline drawing of electron tube type DOD-042.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review Activities:
Army - CR4
Navy - AS, CG, OS, SH
Air Force - 17, 19, 84

Preparing Activity:
DLA - CC

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